ETAS RTA-CAR 12.7.0

RTA-CAR Security Case Summary

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Contents

1 Preface 4

2 Acronyms 5

3 Product Overview 6

4 Cybersecurity Process 7

4.1 Customer Cybersecurity Agreement (SEP-S3) 7

4.2 Supplier Cybersecurity Agreement (SEP-S4) 7

4.3 Cybersecurity Requirements Interface (SEP-S5) 7

4.4 Cybersecurity Threat and Risk Analysis (SEP-S6) 8

4.5 Cybersecurity Concept (SEP-S7) 8

4.6 Secure Implementation (SEP-S8) 8

4.7 Cybersecurity Verification and Validation (SEP-S9) 8

4.8 Security Monitoring (SEP-S10) 8

4.9 Security Vulnerability Management (SEP-S11) 9

4.10 Incident Management (SEP-S12) 9

4.11 Security Changes (SEP-S13) 9

4.12 Decommissioning (SEP-S14) 9

5 Appendices 11

5.1 Security Assumptions 11

5.2 Recommendations for UN R155 Annexure 5 11

6 ETAS Contact Addresses 15

6.1 Technical Support 15

6.2 ETAS HQ 15

7 Document History 16

1 Preface

This document provides an overview of the *ISO/SAE 21434* Cybersecurity Case of the RTA-CAR product.

2 Acronyms

Term	Definition
BSW	Basic Software
CIA	Cybersecurity Interface Agreement
CSMS	Cyber Security Management System
CVSS	Common Vulnerability Scoring System
HSM	Hardware Security Module
NVD	National Vulnerability Database
OEM	Original Equipment Manufacturer
OSS	Open-Source Software
SBOM	Software Bill Of Material
SEP	Security Engineering Process
TARA	Threat And Risk Analysis

3 Product Overview

RTA-CAR is the ETAS integrated environment to configure and generate the Classic AUTOSAR stack for ECUs.

The RTA-CAR solution includes the following products:

- ISOLAR-A: AUTOSAR Authoring Tool that assists users in designing application software to AUTOSAR standards.
- ISOLAR-B: BSW Configuration Tool.
- RTA-BSW: Basic Software automatic configuration and code generation.
- RTA-RTE: Run-time Environment Generator.
- RTA-OS: Operating System Configurator and Generator.
- RTA-A2L: A2L Generator for use with measurement and calibration tools.
- MemMapGen: Memory Map Generator.

4 Cybersecurity Process

RTA-CAR is developed according to ETAS's CSMS (Cybersecurity Management System) and SEP (Security Engineering Process). This includes:

1. RTA-CAR desktop tools (tools and code generators installed on the user's Computer).
2. RTA-CAR generated code (C code integrated into ECUs).

The figure below shows a high-level overview of the ETAS SEP process (Figure 4.1):

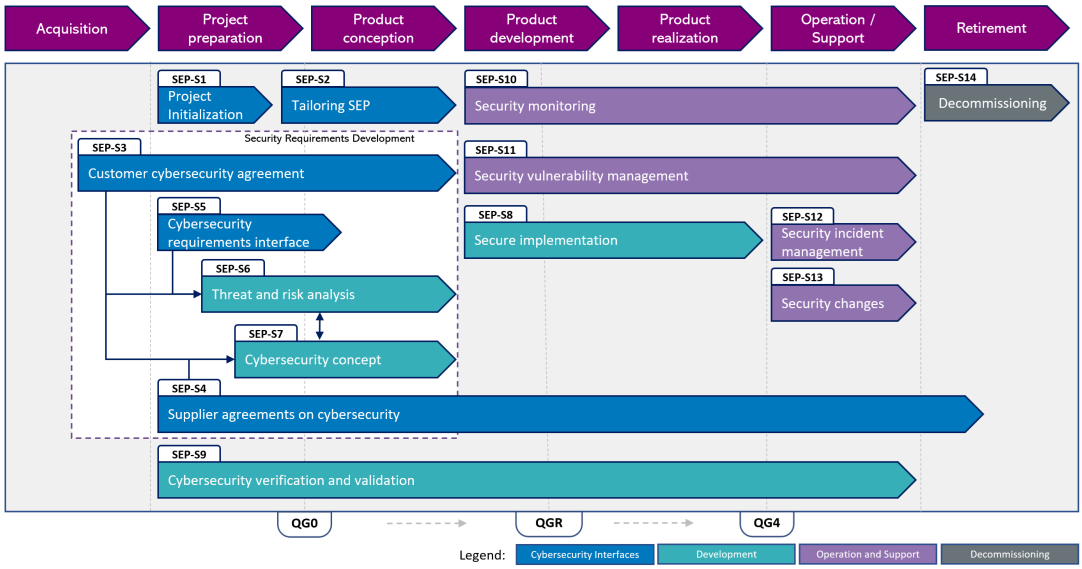

Figure 4.1.ETAS Security Engineering process

> **NOTE**
>
> Relevant sections of the SEP are described in further detail in this document.

The ETAS CSMS, including the SEP, has been assessed by independent German auditor TÜV Süd. The certificate is available here.

4.1 Customer Cybersecurity Agreement (SEP-S3)

A Cybersecurity Interface Agreement (CIA), per *ISO/SAE 21434*, can be agreed upon request.

4.2 Supplier Cybersecurity Agreement (SEP-S4)

RTA-CAR does not include components from third-party suppliers.

4.3 Cybersecurity Requirements Interface (SEP-S5)

Cybersecurity product requirements, including those from customers, are

handled within the product's Standard Requirement Engineering process. Assumptions and claims resulting from the development of these requirements are listed in Security Assumptions (see Section 5.1).

4.4 Cybersecurity Threat and Risk Analysis (SEP-S6)

A TARA was conducted for RTA-CAR and informs the Security Assumptions (see Section 5.1) and Cybersecurity Concept (see Section 4.5).

4.5 Cybersecurity Concept (SEP-S7)

A cybersecurity concept in the context of *ISO/SAE 21434*, either on the vehicle or ECU level, is project specific. RTA-CAR provides a set of tools to generate code, which implements security features. These features can, in turn, be used by the projects to realize their security concept.

To demonstrate how these security features may be used as part of a security concept, the *UN Regulations No 155 (Annexure 5)* document is taken as a reference. Security measures/recommendations (based on this Annexure) are provided in Recommendations for UN R155 Annexure 5 (see Section 5.2).

> **NOTE**
>
> These are intended as *recommended* security features; the Project must choose features and a subsequent configuration based on its own TARA.

4.6 Secure Implementation (SEP-S8)

RTA-CAR generated code complies with *MISRA C:2012* and *CERT-C L1*. The product documentation includes compliance statements covering this in more detail.

4.7 Cybersecurity Verification and Validation (SEP-S9)

As described in the Cybersecurity Concept (see Section 4.5), it is assumed that the integration of RTA-CAR generated code follows *ISO/SAE 21434*, or a similar cybersecurity management process, for any security relevant project.

Features such as those detailed in Recommendations for UN R155 Annexure 5 (see Section 5.2) are designed to fulfil project security requirements. ETAS takes measures, such as those described in the Secure Implementation (see Section 4.6), to reduce the chance of vulnerabilities in these features.

Projects integrating RTA-CAR generated code must perform cybersecurity verification and validation of the integrated ECU containing both the generated code and the RTA-CAR (delivered) embedded code.

4.8 Security Monitoring (SEP-S10)

The ETAS CSMS includes the monitoring of potential vulnerabilities and risks. The following channels are continuously monitored for security vulnerabilities:

1. Requests raised by customers via the product support channels.

2. Product Security Incidence Response Team (PSIRT) which includes the monitoring of:
   - Security news websites, e.g. BleepingComputer, heise, Golem, Hacker News.
   - Social networks, e.g. Twitter/X.
   - Threat information sources, e.g. Malwarebytes Threat Intelligence, Google.
   - Project Zero.
   - Auto-ISAC.
3. SBOM-based monitoring of dependencies (including OSS) against vulnerabilities published by multiple sources (including e.g. NVD).

4.9 Security Vulnerability Management (SEP-S11)

The Vulnerability Management process is triggered in response to vulnerabilities found by the Security Monitoring (see Section 4.8). The Common Vulnerability Scoring System (CVSS) recommended by *ISO/SAE 21434* is used. This includes:

1. Analysis to confirm the existence of the vulnerability, threat scenarios and likelihood. A CVSS score is assigned based on this analysis.
2. For a CVSS score of *Medium or Greater*, the vulnerability is reported to customers, and a fix is created and delivered.
3. For all CVSS scores/ratings, the vulnerability is fixed in the latest version of RTA-CAR.
4. If applicable, the TARA is updated.

4.10 Incident Management (SEP-S12)

RTA-CAR manages security incidents through the Security Monitoring (see Section 4.8) and Security Vulnerability Management (see Section 4.9) processes.

4.11 Security Changes (SEP-S13)

Product updates, including those affecting security features, follow the standard RTA-CAR development processes. These include requirements management, testing and development, resulting in new product versions including updates to documentation e.g. the User Guides and Release Notes.

4.12 Decommissioning (SEP-S14)

Should ETAS decide to discontinue RTA-CAR, a defined process would be followed that includes the following customer notifications:

1. An announcement of the decision to discontinue RTA-CAR, including a timeline for the end of maintenance.
2. End Sales.

3. End Maintenance.

5 Appendices

5.1 Security Assumptions

1. The IT environments in which the RTA-CAR tools are used (in which, therefore, the generated code is managed, integrated and deployed) are trusted.
2. Persons using the RTA-CAR tools (who are, therefore, involved in the management, integration and deployment of the generated code) are trusted.
3. RTA-CAR users follow guidance in the product documentation.
4. The ECU where RTA-CAR generated code is deployed is trusted. This means the ECU development follows an *ISO/SAE 21434*-compliant Security Engineering process.

5.2 Recommendations for UN R155 Annexure 5

Threats to *Vehicle communication channels*	Mitigation Reference	Mitigation	Recommended RTA-CAR Security Controls
Spoofing of messages (e.g. 802.11p V2X during platooning, GNSS messages, etc.) by impersonation.	M10	The vehicle shall verify the authenticity and integrity of messages it receives.	Crypto stack + CycurHSM/CycurSoC using Signature Verification and Hashing mechanisms.
Sybil attack (in order to spoof other vehicles as if there are many vehicles on the road).	M11	Security controls shall be implemented for storing cryptographic keys (e.g. use of Hardware Security Modules).	CycurHSM/CycurSOC or any Hardware Root of Trust is integrated with RTA-CAR and used to store secrets like cryptographic keys.
Communication channels permit code injection into vehicle held data/code e.g. tampered software binary might be injected into the communication stream.	M10, M6	The vehicle shall verify the authenticity and integrity of messages it receives. Systems shall implement security by design to minimize risks.	RTA-CAR Crypto stack + CycurHSM /CycurSoC using Signature Verification and Hashing mechanisms. SecOC, TLS, IpSec, MACSEC for channel authentication.
Communication channels permit manipulation of vehicle-held data/code.	M7	Access control techniques and designs shall be applied to protect system data/code.	Secure Diagnostics.
Accepting information from an unreliable or untrusted source.	M10	The vehicle shall verify the authenticity and integrity of messages it receives.	SecOC (with Freshness management), TLS, IpSec, MACSEC, Crypto stack for channel authentication.
Man-in-the-middle attack/session hijacking.	M10	The vehicle shall verify the authenticity and integrity of messages it receives.	SecOC (with Freshness management), TLS, IpSec, MACSEC, Crypto stack for channel authentication.

Replay attack e.g. an attack against a communication gateway allows the attacker to downgrade software of an ECU or firmware of the gateway.	M10	The vehicle shall verify the authenticity and integrity of messages it receives.	SecOC (with Freshness management), TLS, IpSec, MACSEC, Crypto stack for channel authentication.
Interception of information/interfering radiations/monitoring communications.	M12	Confidential data transmitted to/from the vehicle shall be protected.	Crypto stack, TLS, IPsec, MACSEC.
Gaining unauthorized access to files or data.	M8	Through system design and access control, unauthorized personnel must not be able to access personal or system-critical data. Example of Security Controls can be found in OWASP.	Secure Diagnostics.
Sending a large number of garbage data to the vehicle information system, so that it is unable to provide services in the normal manner.	M13	Measures to detect and recover from a denial-of-service attack shall be employed.	Intrusion Detection (IDSM), Firewall (CycurGate).
Black-hole attack, disruption of communication between vehicles by blocking the transfer of messages to other vehicles.	M13	Measures to detect and recover from a denial-of-service attack shall be employed.	Intrusion Detection (IDSM), Firewall (CycurGate).
An unprivileged user is able to gain privileged access, for example root access.	M9	Measures to prevent and detect unauthorized access shall be employed.	Secure Diagnostics.
Virus embedded in communication media infects vehicle systems.	M14	Measures to protect systems against embedded viruses/malware should be considered.	Secure Boot (not an RTA-CAR product feature) but enabled by HSM e.g. CycurHSM/CycurSOC.
Malicious internal (e.g. CAN) messages.	M15	Measures to detect malicious internal messages or activity should be considered.	SecOC.

Malicious V2X messages, e.g. infrastructure to vehicle or vehicle-vehicle messages (e.g. CAM, DENM).	M10	The vehicle shall verify the authenticity and integrity of messages it receives.	Crypto stack + CycurHSM/CycurSoC using Signature Verification and Hashing mechanisms. SecOC (with Freshness management), TLS, IPsec, MACSEC.
Malicious diagnostic messages.	M10	The vehicle shall verify the authenticity and integrity of messages it receives.	Crypto stack + CycurHSM/CycurSoC using Signature Verification and Hashing mechanisms. SecOC (with Freshness management), TLS, IPsec, MACSEC.
Malicious proprietary messages (e.g. those normally sent from OEM or component/system /function supplier).	M10	The vehicle shall verify the authenticity and integrity of messages it receives.	Crypto stack + CycurHSM/CycurSoC using Signature Verification and Hashing mechanisms. SecOC (with Freshness management), TLS, IPsec, MACSEC.

6 ETAS Contact Addresses

6.1 Technical Support

Technical support is available to all users with a valid support contract. If you do not have a valid support contract, please contact your regional sales office.

- https://www.etas.com/en/contact.php

The best way to get technical support is using the online support portal RTA Hotline:

- https://rtahotline.etas.com/

It is helpful if you can provide technical support with the following information (where relevant):

- Your support contract number
- The version of the ETAS tools you are using
- The version of the compiler tool chain you are using
- The command line (or reproduction of steps) that result in an error message
- The error messages or return codes you received (if any)
- Your RTA-CAR Project or if not possible the arxml file(s) containing the error(s).
- The *Diagnostic.dmp* if it was generated

6.2 ETAS HQ

ETAS GmbH Borsigstraße 24 70469 Stuttgart Germany	Phone:	+49 711 3423-0
	Fax:	+49 711 3423-2106
	WWW:	www.etas.com

7 Document History

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0.1	2024-07-04	For review	Initial Version	Arup Mukherji (ETAS-VOS /ENG-ARC)
1.0	2024-07-26	Released Reviewed	Document marked Internal	Arup Mukherji (ETAS-VOS /ENG-ARC)